

Profile

Computational neuroscientist and research lead working across experimental neuroscience and machine learning. My graduate research examined social decision-making in freely moving macaques using hierarchical Bayesian models and inverse reinforcement learning. At Netholabs I built a multimodal human research programme and helped develop large-scale systems for animal behaviour. My current interests centre on olfaction as an interface between environmental exposure and neurodegenerative disease, and on the safety of AI systems that act in embodied and biological settings.

Skills

Programming	Python; MATLAB; R; Git; Linux; scientific computing and reproducible analysis
Machine learning	PyTorch; transformers; VQ-VAE; diffusion models; representation learning; multimodal prediction and model evaluation
Quantitative methods	Hierarchical Bayesian inference in brms; multinomial models; reinforcement learning; MaxEnt inverse reinforcement learning; Markov decision processes; canonical correlation analysis
Computer vision	DeepLabCut; SLEAP; Anipose; multi-view pose estimation; camera calibration; 3D reconstruction and kinematic analysis
Neural and physiological	Awake 7T fMRI; EEG; fNIRS; eye tracking; ECG and HRV; respiration; EMG; inertial sensing and motion capture
Experimental systems	LSL/XDF synchronisation; multimodal protocol design; data schemas; quality control; camera systems; CAD; 3D printing and rapid prototyping
Wet laboratory	Fixed neural tissue; cryosectioning; histology; immunohistochemistry; staining; immunolabelling and microscopy
Human research	Experimental design; ethics and consent; participant operations; data governance; scientific communication

Experience

Head of Research, Human Programme

Netholabs

Mar–Aug 2026

London, UK

- Joined as the second employee in January 2026 and became Head of Research in March as the organisation grew to approximately 16–20 people.
- Built the human research programme from initial study questions through protocols and participant operations. Established data-governance and quality-control processes.
- Designed synchronised acquisition for Kernel Flow 2 fNIRS and EEG alongside eye tracking. Connected physiological sensors and motion capture through Lab Streaming Layer and XDF.
- Set targets for 12 hours of stationary recordings from four participants and 24 hours of mobile recording. Coordinated research delivery across hardware and modelling teams.

Research and Engineering, Animal Behaviour and Robotics

Netholabs

Jan–Aug 2026

London, UK

- Helped scale continuous mouse-behaviour capture from three cages producing approximately 375 million time points towards a ten-cage system.
- Evaluated VQ-VAE representations of 100 ms behavioural motifs and diagnosed codebook collapse. In one model codebook perplexity tracked motion energy at $\rho = 0.827$.
- Contributed to a 100M-parameter Behaviour GPT benchmark and a 152M-parameter multimodal model. Worked with diffusion models that generated four-second trajectories.
- Transferred generated mouse movement into a real-time Unitree G1 goalkeeping task. Also contributed to a LeRobot system for electrical circuit assembly.

Graduate Researcher, Integrated Program in Neuroscience

McGill University / RIKEN Center for Brain Science

2023–2026

Canada and Japan

- Designed a 3D two-armed-bandit social foraging task for six Japanese macaques. Collected 1,730 trials across 88 blocks in solo and social contexts.
- Developed hierarchical multinomial models in brms to estimate how value and social context changed exploration. Formalised the task as a Markov decision process and developed MaxEnt inverse-reinforcement-learning analyses.
- Designed passive-viewing fMRI experiments on optimal and suboptimal choice from first-person and third-person perspectives.
- Worked on awake 7T macaque and marmoset fMRI with concurrent eye tracking and physiological acquisition. Developed a marmoset paradigm crossing social identity with visual perspective.

RIKEN International Program Fellow

RIKEN Center for Brain Science

2024–2025

Tokyo, Japan

- Developed reproducible acquisition and analysis workflows for non-human-primate studies that linked behaviour with physiological state. Participated in the RIKEN CBS Training Program.

Research Intern, Histology and Immunohistochemistry

International Center for Primate Brain Research

Sep–Oct 2024

Shanghai, China

- Completed a seven-day primate neurobiology school followed by a one-month laboratory placement with Prof. Henry Evrard. Processed more than 75 macaque brain samples through cryosectioning and staining. Performed immunolabelling and microscopy

with careful anatomical documentation.

Research Assistant

Royal Holloway, University of London

2022–2023

London, UK

- Designed and ran studies using virtual reality and eye tracking to examine stress and spatial memory. Managed participant testing and reproducible analysis workflows.

Research and Podcast Intern

Plasticity Lab, University of Cambridge

2026

Cambridge / Remote

- Research support for a programme on sensory substitution and augmentation. Developed the smell domain through literature research and expert mapping with attention to olfactory diagnosis and trigeminal substitution.

Neuroscience Consultant

NeuroZen

2026

Remote

- Designed and analysed a controlled EEG study of a neurostimulation device. Selected outcome measures and interpreted pilot data before specifying a powered follow-up study with a sham condition.

Neuroscience Public Communication Consultant

Voltage TV

2026

London, UK

- Translated research on adolescent cognition and brain development into accessible material for television development.

Python Tutor and Workshop Lead

Freelance / Imperial College London

2025–present

London / Remote

- Teach Python for scientific computing and reproducible research. Designed and delivered a practical Python workshop for NHS staff at Imperial College London in February 2026.

Academic Language Editorial Assistant

Freelance

2021–present

Remote

- Edited and translated more than 30 journal articles and theses. Work has also included grant proposals for multilingual researchers.

Language and Culture Assistant

NALCAP, Spanish Ministry of Education

2021–2022

Granada, Spain

- Taught English in public primary schools and developed classroom materials. Supported cultural exchange activities for Ukrainian refugees.

Mediator and Interpreter

Caritas Migrant Center

2020

Nicosia, Cyprus

- Provided interpretation and psychosocial mediation for asylum seekers in psychiatric and legal settings.

Selected projects

- Olfactory route to neurodegeneration.** Developing a research programme on whether inhaled toxicants can initiate epithelial injury and innate immune activation before neuroinflammation or protein pathology appears along the olfactory route. Proposed work connects post-mortem olfactory bulb tissue with living mucosal models and computational pathology.
- Auracle / Canary.** Co-developed a nose-worn system for VOC sensing and personal filtration, linking short-timescale exposure measurement to olfactory transport into the nervous system. Winner of the Cambridge NeuroWorks Project Motherboard hackathon.
- Macaque social foraging.** Modelled exploration under changing social context using a naturalistic task with known reward values. Developed private and social learning models with hierarchical Bayesian inference and single-agent or multi-agent IRL.
- Multimodal behavioural foundation models.** Worked on tokenisation and evaluation for autoregressive models of mouse behaviour. Examined whether latent codes captured meaningful motifs or simple movement intensity.
- Dual-EEG interpersonal synchrony.** Built an eight-channel SURFER dual-EEG system and used canonical correlation analysis to measure shared neural dynamics during natural interaction. Winner of a Neurosphere natHACKS award.
- Robot electrician.** Contributed to a machine-vision and imitation-learning system for electrical circuit assembly at the Hugging Face LeRobot Global Hackathon.
- M.Oyster.** Developed a biomimetic material concept inspired by oyster-shell formation and repair. Winner of the Nucleate MaterialHack Blue Sky Award.
- VR neurotechnology.** Built projects including Tipsea Dipsea for VR Stroop testing and Monkey Sea for naturalistic behaviour. Also worked on Monkey Doom to explore breathing and interpersonal space.
- Parkinson's technology.** Developed an EMG-controlled exoskeleton project through the McGill Biomechanics Society. Worked on VR paradigms for movement and cognition in Parkinson's disease.
- AI safety and digital minds.** Mentored the Apart Research Digital Minds Sprint and participated in Neuromatch AI Sentience. Current work examines objective inference and the limits of interpreting behaviour without access to ground-truth goals.

Education

Graduate study, Integrated Program in Neuroscience

McGill University

2023–2026

Coursework completed, GPA 4.0/4.0

MSc Applied Neuroscience

Royal Holloway, University of London

2022–2023

Distinction

Selected training: Cambridge NeuroWorks What If Programme; Crick Innovation Challenge; Apart Research Digital Minds Sprint mentor; Neuromatch AI Sentience programme (2026). Le Wagon Data Science and Machine Learning bootcamp, London, three-month intensive; Neurohackademy, University of Washington eScience Institute; Women in Science Machine Learning Programme, Tokyo (2025). RIKEN CBS Training Program (2024–25). ICPBR Summer School and laboratory internship, Shanghai (2024). RIKEN CBS Learning and Imagination Summer Program (2023). Methods and Statistics in Social Sciences, University of Amsterdam, Grade A; Video Game Design and Design Principles, Rhode Island School of Design, Grade A (2022).

Publications and written research

- Leyser, H. F. and Cooper, K. Motion, Mirror Neurons and the Human–Robot Relationship. In revision. Equal primary contribution to the conceptual framework and neuroscience research.
- Leyser, H. F. Why Mirror Neurons Are the Most Important Finding in Psychology. Winner of the Royal Holloway Postgraduate Student Essay Competition, 2023.
- Leyser, H. F. High Stakes: The Effect of Depth-Induced Stress on Memory Encoding Strategies. MSc thesis, Royal Holloway, 2023.
- Leyser, H. F. Behind Barbed Wires: Internment and Belonging in 1940s Britain. BA thesis, University of Oxford, 2021.

Presentations

Oral presentations

- Sep 2024. International Center for Primate Brain Research Summer School, Shanghai. “Sociability, Hierarchy, and Reward: A Novel Decision-Making Framework.”
- May 2024. QBIN Scientific Day, Montréal. “Integrating Movement and Decision-Making: An Embodied Cross-Species Approach.”
- Apr 2024. British Neuroscience Association Members Meeting. “Embodied Foraging: Explore vs Exploit Social Decision-Making with Japanese Macaques.”
- Mar 2024. Three Minute Thesis Competition, Montréal. “Fight or Flight Mechanisms: Implications for Police Brutality and Beyond.”
- Aug 2023. RIKEN CBS Summer Programme Review, Tokyo. “Monkey Business: The Role of Sociability and Hierarchy on Exploration and Exploitation Trade-offs.”

Posters

- Jan 2025. Co-Creation International Conference, Tokyo. “InterAction: Neural Mechanisms of Social Behaviour.”
- Nov 2024. RIKEN CBS Retreat, Gotemba. “The Social Brain in Action.”
- Sep 2024. International Center for Primate Brain Research, Shanghai. “Multi-Agent Movement and Decision-Making Dynamics.”
- Jul 2024. Japanese Neuroscience Society, Fukuoka. “Embodied Decision-Making in Non-Human Primates: A Social Foraging Paradigm Using a 3D Two-Armed Bandit Task.”
- May 2024. Neuropsychology Day Symposium, Montréal. “Cross-Species Investigation of the Neurocognitive Dynamics of Explore/Exploit Decision-Making.”
- Feb 2024. COSYNE, Lisbon. “Unveiling Movement Patterns as Cognitive Nodes: Exploration and Exploitation in Embodied Decision-Making in Macaques.”
- Jul 2023. RIKEN CBS Lecture Series, Tokyo. “Exploring Multisensory Integration Networks Across the Neurodevelopment of Marmosets.”
- Jun 2023. Generative Episodic Memory Conference, Bochum. “High Stakes: The Effect of Stress on Egocentric and Allocentric Frames of Reference in Memory.”

Invited and internal talks

- Mar 2024. Healthy Brains, Healthy Lives Meeting, Montréal. “Unveiling Movement Patterns as Cognitive Nodes.”
- Mar 2024. Zatorre Lab Journal Club, Montréal. “Rhythm and Syncopation: Movement as Social Communication.”
- Feb 2024. Integrate and Fire Seminar Series, McGill. “Untangling Sex and Gender in Neuroscience.”
- Jan 2024. Non-Human Primates Journal Club, McGill. “Behavioural Acquisition and Deep Learning of Explore/Exploit Paradigms with Macaques.”
- Nov 2023. Integrate and Fire Seminar Series, McGill. “Connected Brains: Why Are We Social?”

Awards and service

2026	Cambridge NeuroWorks What If Fellowship funded by ARIA; Project Motherboard Hackathon Winner for Auracle; Nucleate MaterialHack Blue Sky Award for M.Oyster
2025	UCSF–RIKEN Center for Brain Science Young Investigator Exchange Travel Award; Japanese Neural Network Society Travel Grant
2024–25	RIKEN International Program Fellowship; McGill University Graduate Mobility Award
2024	Neurosphere natHACKS Award; Quebec Bio-Imaging Network Travel for Training Award; International Center for Primate Brain Research full internship award; GREAT Travel Award, McGill University; COSYNE New Attendee Travel Grant
2023–27	McGill University International Differential Fee Waiver and stipend award
2023	McGill Integrated Program in Neuroscience Recruitment Award; Royal Holloway Postgraduate Student Essay Competition Winner; Women in Memory Research Award, Ruhr University Bochum; RIKEN Center for Brain Science travel and accommodation award
2022	StoryFutures Academy Virtual Production Laboratory
Service	Member of the COSYNE Diversity and Inclusion Committee, 2024–present. Vice President of McGill Care Through Tech and Senior Project Advisor to the McGill Biomechanics Society, 2023–2024. Ambassador for Thinking About Thinking and Cambridge NeuroWorks Ambassador, 2026. Speaker and judge at the Imperial College Neurotechnology Hackathon, February 2026.

Languages: English and German (bilingual); French and Spanish (full professional proficiency); Arabic (basic); Latin and Ancient Greek